

Network Diagnostic Tools

During my network testing, I found that **ping** is the most useful tool for checking whether a device is reachable. When I ping an offline device on my local network, it shows "**Request timed out,**" which indicates that the device is not connected or cannot be reached.

For a **website that is not loading**, **nslookup** is the best tool because it checks whether the domain name can be translated into an IP address. In my test, `nslookup www.zomato.com` successfully returned the IP addresses **104.18.42.183** and **172.64.145.73**, confirming that DNS resolution was working.

To diagnose a **slow connection**, **tracert** is the most useful tool because it shows every network hop between my computer and the destination. In my test to www.flipkart.com, the destination was reached at **Hop 15**, while several intermediate hops showed "**Request timed out,**" helping identify where delays or blocked responses might occur.

Overall, **ping** is best for checking connectivity, **tracert** is best for finding delays along the network path, and **nslookup** is best for troubleshooting DNS-related problems.